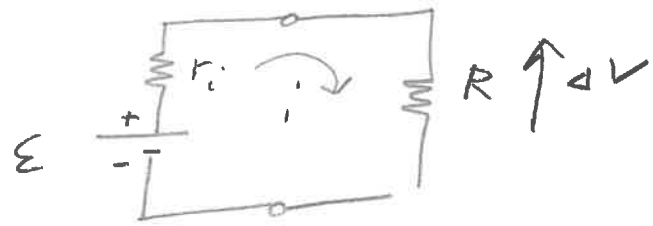
Chpt 28 problems

1.



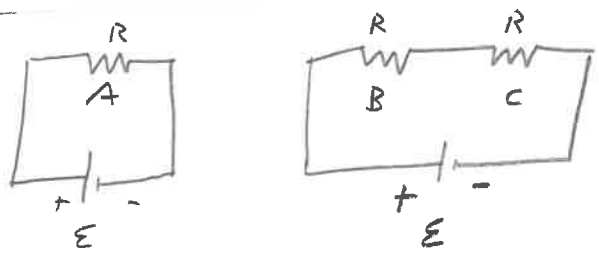
$\mathcal{E} = 12.6\ V$
$r_i = 0.056\ \Omega$
$R = 4.60\ \Omega$ (headlights)

$$i = \frac{\mathcal{E}}{r_i + R}$$ since $r_i + R$ = total resistance

a) $\Delta V = iR = \mathcal{E}\left(\frac{R}{r_i + R}\right) = \boxed{12.4 \text{ volts}}$

b) If $i = 35\ A$ is supplied by starter motor, then
$\Delta V = iR = 35A \times 4.60\Omega = \boxed{161 \text{ volts}}$

2.



a) $I_A = \boxed{\frac{\mathcal{E}}{R}}$

$I_B = \boxed{\frac{\mathcal{E}}{2R}}$, $I_C = \boxed{\frac{\mathcal{E}}{2R}}$

b) Brightness is proportional to power dissipated in each light bulb.

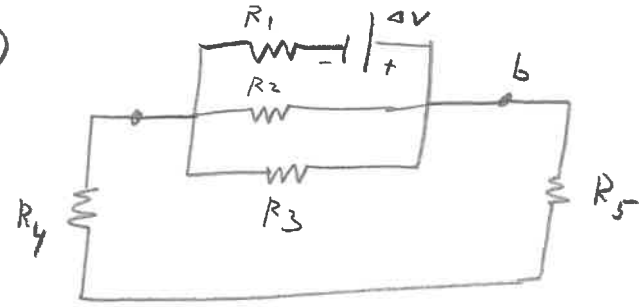
$P_A = I_A^2 R$
$P_B = I_B^2 R$
$P_C = I_C^2 R$

$P_B = P_C$ since $I_B = I_C$
so same brightness $\boxed{=}$

c) Note that $I_B = I_C = \frac{1}{2} I_A$
Thus, $P_B = \left(\frac{1}{2} I_A\right)^2 R = \frac{1}{4} I_A^2 R = \frac{1}{4} P_A$

so $\boxed{P_A = 4P_B}$ (Brightness of A > Brightness of B) (or C)

3



$\Delta V = 25\ V$
$R_1 = R_2 = 10\Omega$
$R_3 = R_4 = 5\Omega$
$R_5 = 18\ \Omega$

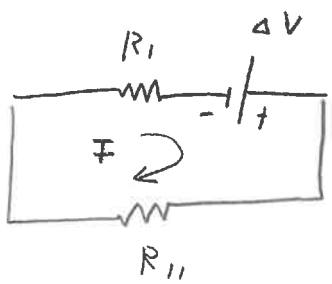
a) Find current through R_5

Combine R_4, R_5 in series $\rightarrow$ $R_{45} = R_4 + R_5 = 23\Omega$

Combine R_2, R_3, R_{45} in // $\rightarrow$ $R_{//} = \dfrac{1}{\frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_{45}}}$

$= \dfrac{1}{\frac{1}{10} + \frac{1}{5} + \frac{1}{23}}$

$= 2.91\ \Omega$



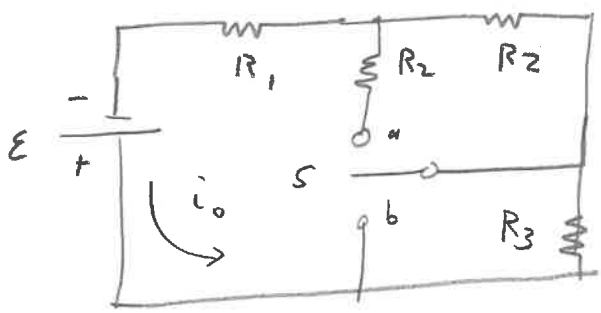
$I = \dfrac{\Delta V}{R_1 + R_{//}} = \dfrac{25\ \text{volts}}{10\Omega + 2.91\Omega}$

$\Delta V_{//} = I R_{//} = 5.64$ volts

$I_5 = \dfrac{\Delta V_{//}}{R_{45}} = \dfrac{5.64\ \text{volts}}{23\ \Omega} = \boxed{0.245\ \text{amps}}$

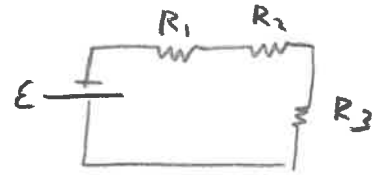
b) $\Delta V_{ab} = \Delta V_{//} = \boxed{5.64\ \text{volts}}$

4.



$\mathcal{E} = 3\ V$

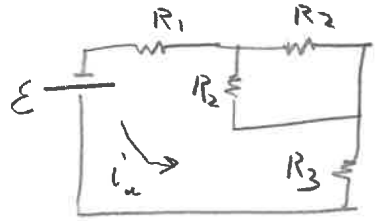
Switch open:



$i_0 = \dfrac{\mathcal{E}}{R_1 + R_2 + R_3}$

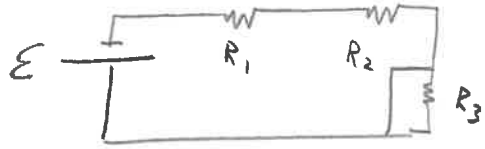
$= 1.09\ mA$

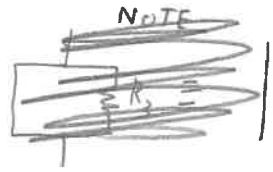
Position a:



$i_a = \dfrac{\mathcal{E}}{R_1 + \frac{R_2}{2} + R_3} = 1.22\ mA$

Position b:





$$i_b = \frac{\mathcal{E}}{R_1 + R_2} = 2.01\ mA$$

Thus,

$$R_1 + R_2 + R_3 = \frac{\mathcal{E}}{i_0} \equiv C_1 = 2752\,\Omega \quad (1)$$

$$R_1 + \frac{R_2}{2} + R_3 = \frac{\mathcal{E}}{i_a} \equiv C_2 = 2459\,\Omega \quad (2)$$

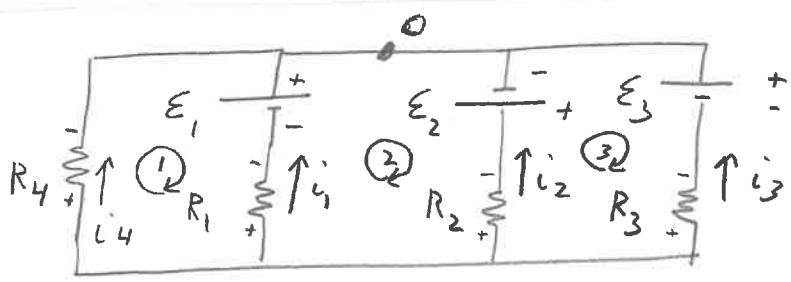
$$R_1 + R_2 = \frac{\mathcal{E}}{i_b} \equiv C_3 = 1493\,\Omega \quad (3)$$

(c) Subtract (3) from (1) $\rightarrow$ $R_3 = C_1 - C_3 = \boxed{1.26\ k\Omega}$

(b) subtract (2) from (1) $\rightarrow$ $\frac{R_2}{2} = C_1 - C_2 \rightarrow R_2 = 2(C_1 - C_2) = \boxed{0.587\ k\Omega}$

(a) Then $R_1 = C_3 - R_2 = \boxed{0.906\ k\Omega}$

5.



$\mathcal{E}_1 = 40\,V$
$\mathcal{E}_2 = 360\,V$
$\mathcal{E}_3 = 80\,V$
$R_1 = 80\,\Omega$
$R_2 = 20\,\Omega$
$R_3 = 70\,\Omega$
$R_4 = 200\,\Omega$

(a) junction rule at 0: $i_1 + i_2 + i_3 + i_4 = 0$ (1)

Loop rule 1: $-i_4 R_4 - \mathcal{E}_1 + i_1 R_1 = 0$ (2)

Loop rule 2: $-i_1 R_1 + \mathcal{E}_1 + \mathcal{E}_2 + i_2 R_2 = 0$ (3)

Loop rule 3: $-i_2 R_2 - \mathcal{E}_2 - \mathcal{E}_3 + i_3 R_3 = 0$ (4)

Convert above 4 equations into a matrix equation.

Matrix Equation

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ R_1 & 0 & 0 & -R_4 \\ -R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 0 \\ \mathcal{E}_1 \\ -\mathcal{E}_1 - \mathcal{E}_2 \\ \mathcal{E}_2 + \mathcal{E}_3 \end{bmatrix}$$

Solving matrix equation

$$M \vec{v} = \vec{w}$$

$\rightarrow$ $\vec{v} = M^{-1} \vec{w}$ when M^{-1} is inverse matrix of M

substituting the values for $\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3$ and R_1, R_2, R_3, R_4

$\rightarrow$

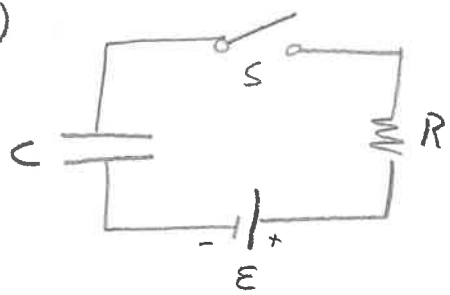
$i_1 = 3$ amps		$(80\,\Omega)$
$i_2 = -8$ amps		$(20\,\Omega)$
$i_3 = 4$ amp		$(70\,\Omega)$
$i_4 = 1$ amps		$(200\,\Omega)$

(b) $\Delta V_3 = -R_3\, i_3$

$= -70\,\Omega \times 4$ Amps

$= \boxed{-280 \text{ volts}}$

(6.)



$R = 3\,M\Omega = 3\times10^{6}\,\Omega$

$C = 6\,\mu F = 6\times10^{-6}\,F$

$\mathcal{E} = 34\,V$

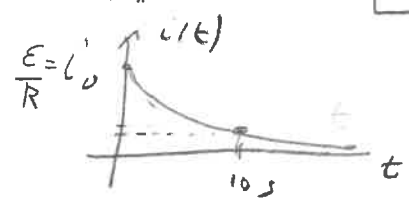
(charging capacitor)

a) $\tau = RC = \boxed{18\,s}$

b) $\Delta V_{max} = \mathcal{E} \rightarrow Q_{max} = C\,\Delta V_{max} = C\mathcal{E} = \boxed{204\,\mu C}$

c) $i(t) = i_0\, e^{-t/RC}$

$= \dfrac{\mathcal{E}}{R}\, e^{-t/RC}$



$i(t=10\,s) = \dfrac{34\,V}{3\times10^{-6}\,\Omega}\, e^{-10\,s/18\,s}$

$= \boxed{6.5\,\mu A}$

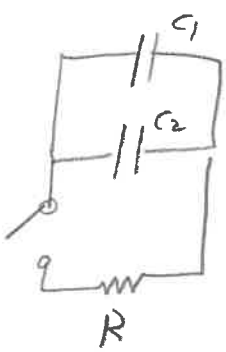
7. Heater : $P_1 = 2\times10^3$ W
Toaster : $P_2 = 500$ W
Grill : $P_3 = 1000$ W

$\Delta V = 120$ V

a) $I_1 = P_1/\Delta V = \boxed{16.67 \text{ amps}}$
$I_2 = P_2/\Delta V = \boxed{4.167 \text{ amps}}$
$I_3 = P_3/\Delta V = \boxed{8.33 \text{ amps}}$

b) $I_{tot} = I_1 + I_2 + I_3 = \boxed{29.16 \text{ amps}} < 30$ amps
so circuit breaker will <u>not</u> be tripped

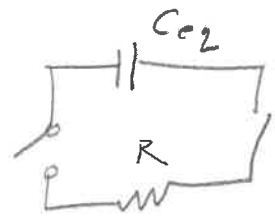
8.



$C_1 = 3\,\mu F$
$C_2 = 2\,\mu F$
$R = 500\,\Omega$

$\Delta V = 24$ volts
(discharging capacitors)

Equivalent circuit



$C_{eq} = C_1 + C_2 = 5\,\mu F$

$Q_{eq} = C_{eq}\,\Delta V = 120\,\mu C$
(total initial charge on C_1 and C_2)

When discharging:

$$q_{eq}(t) = Q_{eq}\, e^{-t/\tau} \quad \text{where } \tau \equiv R\,C_{eq} = 2.5 \text{ ms}$$

when $t = 3$ ms, $q_{eq}(t=3\text{ ms}) = 61.44\,\mu C$

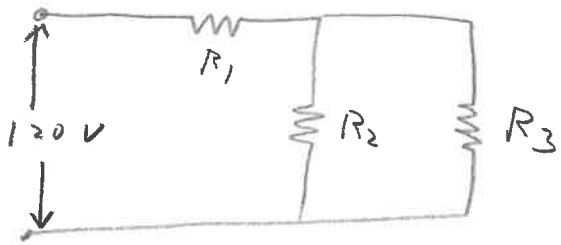
$$\Delta V(t=3\text{ms}) = \frac{q_{eq}(t=3\text{ms})}{C_{eq}} = 12.3 \text{ volts}$$

a) Thus, $Q_1(t=3\text{ms}) = C_1\,\Delta V(t=3\text{ms}) = \boxed{36.9\,\mu C}$

b) $Q(t=3\text{ms}) = C_2\,\Delta V(t=3\text{ms}) = \boxed{24.6\,\mu C}$

c) $i(t=3\text{ms}) = \dfrac{\Delta V(t=3\text{ms})}{R} = \boxed{24.6 \text{ mA}}$

9.



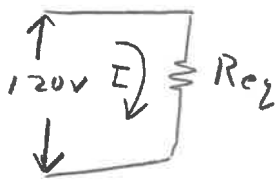
$R_1 = R_2 = R_3 = R$

$P = 45\ W$

$\Delta V = 120\ V$

Recall: $P = \frac{(\Delta V)^2}{R}$ $\rightarrow$ $R = \frac{(\Delta V)^2}{P} = \frac{(120V)^2}{45\ W} = 320\Omega$

a) Equivalent circuit



where $R_{eq} = R_1 + R_{23}$

$R_{23} = \frac{1}{\frac{1}{R_2} + \frac{1}{R_3}} = \frac{1}{\frac{1}{R} + \frac{1}{R}} = \frac{R}{2}$

$\rightarrow$ $R_{eq} = R + \frac{R}{2} = \frac{3R}{2}$

Total power supplied by source:

$P_{tot} = \Delta V \cdot I = \Delta V \cdot \left(\frac{\Delta V}{R_{eq}}\right) = \frac{(\Delta V)^2}{\frac{3R}{2}} = \frac{2}{3} P$

$= \boxed{30\ \text{watts}}$

(Equivalently, $I = \left(\frac{\Delta V}{R_{eq}}\right) = \frac{\Delta V}{\left(\frac{3R}{2}\right)} = \frac{2}{3}\left(\frac{\Delta V}{R}\right) = 0.25A$

$\rightarrow$ $P_{tot} = \Delta V \cdot I = 120V\,(0.25A) = 30\ \text{watts}$)

b) $\Delta V_1 = R_1 I = R I = \boxed{80\ \text{volts}}$

$\Delta V_2 = \Delta V_3 = R_{23} \cdot I = \frac{R}{2} I = \boxed{40\ \text{volts}}$